Ex: 12 Echo client - Server using UDP

13) 11/9/2026

Aim:

To implement an Echo-client server application using UDP Sockets and verify Communication b/w client and Server.

Procedure:

1. Creat a UDP Echo Server using Datagram Socket and Datagram packet
2. Assign a port number to the UDP Server.
3. Compile and execute the UDP Server program
4. Creat a UDP Echo Client using Datagram Socket and Datagram Packet
5. Specify the Server address and port number.
6. Enter a message through the Client
7. The Client Sends the message as a
8. The Server recives the datagram and Sends the received message back.
9. Display the received echoed message at the Client.

Result:

Thus, the UDP Echo Client-Server application was Successfully Implemented, and the Client Successfully reaved echoed message from the Server.

```
Problems  Output  Debug Console  Terminal  Ports  Jupyter
PS C:\Users\Luei\Downloads\CN_EXPT_12> javac UDPEchoServer.java UDPEchoClient.java
PS C:\Users\Luei\Downloads\CN_EXPT_12> java UDPEchoServer
UDP Echo Server listening on port 6000
Received from /127.0.0.1:63830 -> Hello
Received from /127.0.0.1:63830 -> I am studying Computer Networks
Received from /127.0.0.1:63830 -> This is for Testing
```

```
Problems  Output  Debug Console  Terminal  Ports  Jupyter
PS C:\Users\Luei\Downloads\CN_EXPT_12> java UDPEchoClient
UDP Echo Client ready to send to localhost:6000
Type messages to send (type 'bye' to quit):
Hello
Server: Echo: Hello
I am studying Computer Networks
Server: Echo: I am studying Computer Networks
This is for Testing
Server: Echo: This is for Testing
```

UDP Echo Server

```
Import Socket
S= Socket.Socket(Socket.AF_INET
Socket.Sock_DGRAM)
S.bind(("Localhost", 5001))
msg, addr = S.recvfrom(1024)
S.Send to (msg, addr)
S Close
```

UDP Echo Client

```
Import Socket
S = Socket.Socket(Socket.AF_INET
Socket.Sock_DGRAM)
S.Send to (b'Hello Sever',
("Localhost", 5001))
msg, addr = S.recvfrom(1024)
Print(msg.decode())
S.Close()
```